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Comparative Performance Evaluation of One-Stage and Two-Stage Object Detectors for Screw Head Detection and Classification in Disassembly Processes

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Abstract

The process of manually detecting and removing screws during disassembly can be time-consuming and demanding. Automated systems utilizing imaging and deep learning can rapidly and accurately identify screws based on features like head shape and size. Different deep learning methods have varying screw detection capabilities. Choosing the right method significantly impacts system effectiveness, robustness and efficiency. Several generic deep learning architectures have been proposed to address this problem. However, there is little guidance on their suitability for specific scenarios. This paper aims to provide qualified guidance for selecting suitable deep learning methods for screw head detection. The proposed approach involves researching existing object detection methods and choosing two representatives – one employing a one-stage approach, the other using two-stage. These selected methods will be implemented identically and their performance evaluated via rigorous analysis and comparison. The dataset comprises synthetic and real images of 6 screw types across 3 products. YOLOv5 was selected for one-stage detectors, Faster R-CNN for two-stage. Results show the one-stage YOLOv5 generally outperforms Faster R-CNN in precision, recall, speed and training times, while Faster R-CNN delivered superior recall with smaller training datasets.

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1. Introduction

The increasing amount of electrical and electronic waste (WEEE) poses a significant problem for the environment. To effectively decrease the amount of WEEE, it is crucial to replace linear product lifecycles with circular lifecycles, wherein valuable components and materials are reused or recycled [9]. A circular lifecycle for these devices is complex, time-consuming, and cost-intensive due to the high demands on the flexibility of dismantling systems [12].

The recycling of electrical waste is still mostly carried out using the coarse “crush-and-separate” method, in which the equipment is crushed and broken down into tiny particles, which are then separated into reusable components by physical-chemical means [3]. However, these methods can be labor-intensive and not efficient. This leads to improve the recycling processes for WEEE to enhance profitability, efficiency and a positive environmental impact. The manual disassem-

bly of WEEE is also time-consuming and requires significant manpower [1]. Therefore, introducing automation and partial automation to this disassembly process can make a substantial difference. Screws are a widely used and standardized fastener. Thus, unscrewing is an essential task in disassembly systems. Manual screw detection and removal in disassembly processes can be time-consuming and labor-intensive. Automated screw head detection enables fast and accurate identification of screws, resulting in higher overall disassembly efficiency and productivity [13]. Automated screw head detection systems use advanced imaging and deep learning techniques to precisely identify and classify screws based on their head shape, size, and other features, ensuring accurate removal and minimizing the risk of damage to the components. The ability to accurately detect and classify screws based on their visual characteristics varies across different deep learning methods. One-stage and two-stage object detection are two main approaches in the field of deep learning for visual object detection. These approaches differ in the way they detect and classify objects in

an image. Two-stage object detectors typically exhibit higher accuracy in comparison to one-stage detectors. However, this advantage comes at the cost of being slower. Despite their relatively slower processing, two-stage detectors prove to be highly effective when it comes to detecting objects in intricate and cluttered scenes, leading to exceptional precision in their results [2]. One-stage detectors are highly suitable for real-time applications that demand swift processing. However, it is important to note that they might compromise on accuracy. The characteristics and performance of each of these types of detectors depend on the conditions of their use case [2]. Despite the fundamental trade-off between speed and accuracy for the two approaches, there lacks a comparative analysis guiding the selection between them given a particular operating context, so a systematic experimental study is still needed.

Most existing works [24], [20], [26], [16], [5] have focused on demonstrating the viability of deep learning for automated screw detection rather than providing qualified guidance on model selection. The existence of many Deep Learning architectures presents a challenge for practitioners to determine which is best suited for particular problem such as screw head detection. In this study we aim to bridge this gap by comparing and analyzing precision-speed trade-offs and evaluating performance factors of two object detectors - YOLOv5 (one-stage) and Faster R-CNN (two-stage) - experimentally trained on the same dataset to offer qualified guidance to practitioners on selecting the appropriate detector based on this

specific use case requirements and constraints. In this paper we provide a contribution to the automation of disassembly systems to reduce costs and enhance the profitability of circular lifecycles for electrical and electronic devices.

The rest of the paper is organized as follows: Section 2 reviews the relevant related work and explores the object detection frameworks; Section 3 describes the materials and methods proposed in the study; Section 4 presents the results obtained from the experimental study and discusses the main findings; Section 5 presents the conclusions and potential future work.

2. Related Work

Due to advances in deep convolutional neural networks and more powerful graphics processors (GPUs), deep learning models are now widely employed in the field of computer vision [4]. The development period of object detection methods can be divided into two main phases: the phase before 2014 and after 2014 [27]. This paper focuses on modern architectures developed after 2014 with deep learning. The classification of object detection models considered in the context of this paper is shown in Fig. 1.

2.1. Two-Stage Detectors

Two-stage object detectors work in two phases. First, they produce potential bounding boxes that may contain objects. These proposed regions are called region proposals. afterwards, a deep neural network extracts features from the proposed re-

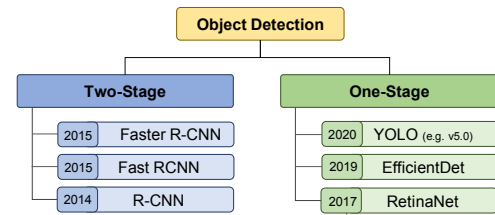


Fig. 1. Classification of general object detection models [11]

gions. In the second stage, a separate classification network analyzes these features. It predicts the class of each object and refines the bounding box coordinates [21]. The characteristics and features of these detectors are explained below:

2.1.1. R-CNN

R-CNN (Region-based Convolutional Neural Network) is an object detection algorithm developed in 2014 by Girshick et al. [7]. It works by first generating region proposals likely to contain objects and then extracting features from these regions using a pre-trained CNN. These features are then used for both classifying the object and refining its bounding box coordinates, resulting in accurate object detection.

2.1.2. Fast R-CNN

Fast R-CNN is an algorithm developed in 2015 by Girshick [6] that builds upon the original R-CNN framework. It integrates the region proposal generation step into the model by introducing a Region Proposal Network (RPN), which shares convolutional operations with the detection network. This reduces redundant computations and speeds up the detection process.

2.1.3. Faster R-CNN

Faster R-CNN (Faster Region-based Convolutional Neural Network) is an advanced algorithm developed in 2015 by Ren et al. [18] that builds upon the R-CNN and Fast R-CNN frameworks. It introduces a Region Proposal Network (RPN) that shares convolutional layers with the detection network, enabling the model to generate region proposals directly from feature maps. These region proposals represent potential regions of interest where objects may be located. The RoI (Region of Interest) pooling step scales these regions to a fixed size for consistent input to subsequent layers. The network then classifies the objects and refines their bounding box coordinates. Faster R-CNN's integration of the RPN leads to an end-to-end trainable system, resulting in significantly improved speed and efficiency in object detection compared to its predecessors.

2.2. One-Stage Detectors

One-stage detectors apply a single neural network to the full image to directly predict objects. The image is divided into a grid of cells, each responsible for detecting objects centered in it. The network generates anchor boxes of varying scales and sizes, scores them for objectness and predicts class probabilities and bounding box refinements. High scoring anchors are

selected after non-maximum suppression filtering, eliminating the need for a separate region proposal step [25]. The characteristics and features of the most popular detectors are described as below:

2.2.1. RetinaNet

RetinaNet is a one-stage object detection model designed to address class imbalance in object detection tasks. It utilizes a deep CNN backbone, extracting features from the input image. Through a Feature Pyramid Network (FPN), it captures features at various scales to detect objects of different sizes. For each spatial location, RetinaNet applies separate heads for object classification and bounding box regression, utilizing anchor boxes as reference frames. The model introduces a focal loss function, prioritizing hard-to-classify examples and improving its ability to learn from challenging cases. This combination of techniques enables RetinaNet to achieve high accuracy in object detection tasks [15].

2.2.2. EfficientDet

EfficientDet is an advanced object detection model based on the EfficientNet architecture, known for its efficiency and high performance. It begins with a deep CNN backbone, typically an EfficientNet, to efficiently extract features from input images. The model employs a Bidirectional Feature Pyramid Network (BiFPN) to aggregate features from different levels, enabling multi-scale object detection. For each feature level, EfficientDet uses separate heads for object classification and bounding box regression. The model also utilizes a variant of the focal loss, focusing more on challenging examples during training to improve its handling of complex cases. These combined techniques make EfficientDet highly efficient and capable of achieving state-of-the-art performance in object detection tasks [22].

2.2.3. YOLO

YOLO (You Only Look Once) is a pioneering object detection model that processes images in a single pass through the neural network. It divides the image into a grid and predicts bounding boxes, objectness scores, and class probabilities for each grid cell simultaneously. This approach makes YOLO significantly faster than traditional methods, making it well-suited for real-time applications. YOLO has evolved through different versions, with each iteration refining its accuracy and speed, and it remains widely used in computer vision tasks [17].

2.3. Screw Head Detection

The field of computer vision and deep learning has seen an increase in interest and progress in object detection [8]. Significant progress has been made in developing and implementing models for automated detection of screw heads [24], [20], [26]. For automatic screw detection and classification, one-stage detector has been applied by Mangold et al. [16]. They developed a vision-based screw detection system using YOLOv5 that can detect and classify 6 different types of screw heads. The results showed achieved high overall precision for screw detection using YOLOv5s, with a mAP@0.5 of 0.984 and mAP@0.5:0.95

of 0.834 on their dataset. Two-stage model was used by Foo et al. [5]. They aimed to develop a vision-based screw detection system using Faster R-CNN with Inception v2 architecture. The results showed higher Precision and recall 91.8% and 83.6% respectively after retraining model on missed detections. A performance comparison between YOLOv5 and Faster R-CNN for screw head detection, based on related work, would not be conclusive. For an adequate comparison, it is necessary to train these models on the same dataset and test them on the same use case.

3. Material and Methods

YOLOv5 and Faster R-CNN were selected as representatives of one-stage and two-stage object detection models respectively for this comparison study. YOLOv5 was chosen as it is one of the most widely used and state of the art one-stage detectors, known for its speed and accuracy trade-offs [10]. On the other hand, Faster R-CNN was chosen as a leading two-stage detector that typifies this approach. As pioneering works in screw detection, YOLOv5 and Faster R-CNN serve as ideal models for comparing one-stage and two-stage methods on this screw detection task. This section presents an overview about the implemented dataset and the used methods in this study.

3.1. Dataset

As a result of the absence of an existing open-source screw head dataset, a new dataset is generated for this study as in [5] and [16]. The dataset contains 2216 images from three different products: monitors, keyboards and CPU cases. These images present screw heads in 6 classes: hex, hex socket, phillips, pozidriv, slotted and torx. The images were manually and automatically generated [19]. 488 images are self-made real images of the specified products and 1728 images are synthetic generated images that aim to supplement real data with expanded variability. Synthetic images were created by CAD modeling and rendering software with intentionally varying state variables, such as orientations, positions, lighting conditions and products' material. This study aims too to evaluate how synthetic data, dataset size and image resolution affect YOLOv5 and Faster R-CNN detector performance. To evaluate the performance of each compared detector, the dataset is organized in 7 distinct training and test configurations as shown in Table 1.

Table 1. Dataset configurations for training.

Nr.	Configuration	Description of subdataset
1	Synthetic	synthetic images
2	Low-Resolution	640x640 pixel images
3	Mid-Resolution	800x800 pixel images
4	Classification	isolated screw head images
5	Small-Size	1064 images
6	Medium-Size	1640 images
7	Complete	2216 images

The multiple configurations systematically vary the number of

training images across real-only, synthetic-only, and combined real and synthetic data. This allows to quantify model performance as dataset size grows. Meanwhile, the images are prepared at three resolutions - 960x960, 800x800 and 640x640 pixels. By evaluating high and low resolution images, it is possible to determine how the information content of the images affects detection performance. Figure 2 provides an example of real and synthetic images from the dataset, illustrating the variability in screw appearance, positioning, and environment between the two data sources.

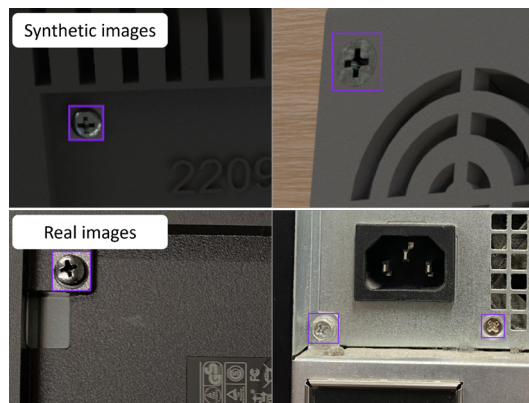


Fig. 2. Synthetic and real image of dataset

3.2. Training

The object detectors were trained and tested on a Ubuntu 18.04.06 operative system with a NVIDIA GeForce RTX 2080 Ti GPU. YOLOv5 was implemented based on the Ultralytics repository [10] and Faster RCNN based on the Detectron2 repository [23]. The training settings are shown in table 2.

Table 2. Training settings.

Object Detector	YOLOv5	Faster R-CNN
Weights	YOLOv5m	Resnet 50 FPN (3x)
Epochs	300	300
Batch size	4	4
Framework	PyTorch	PyTorch

YOLOv5 and Faster R-CNN have multiple adaptable hyperparameters. In this comparison the learning rate parameter was set to 0.01 while the other hyperparameters were chosen as the default values in [10] and [23]. Each object detector will be then fine-tuned with each training configuration of table 1. This results in 7 versions of trained models per object detector.

4. Results and Discussion

To comprehensively evaluate and compare YOLOv5 and Faster R-CNN detection performance, precision and recall metrics are computed using the COCO object detection evaluation protocol. Specifically, $AP_{0.50:0.95}$ and AP_{50} averages precisions

across IoU thresholds from 0.50 to 0.95 [14]. This provides a holistic view of precision over different bounding box matching criteria. Meanwhile, recall metrics assess object detection completeness overall and per class. In addition to detection accuracy, inference speed in FPS is reported to quantify real-time capabilities. As automated screw head detection applications demand fast processing times, this speed benchmarking identifies trade-offs between precision and speed. Finally, training time is compared as an indicator of computational efficiency during model development. Together, these metrics of precision, recall, inference speed and training time provide multi-dimensional insights into the detection proficiency, runtime viability, and training costs of YOLOv5 versus Faster R-CNN. Analyzing these factors by dataset configuration spotlights the advantages and disadvantages of each model under varying data conditions.

4.1. Trained with synthetic data

Configuration 1 (see table 1) provides initial insight into model performance using purely synthetic training data generated under ideal, controlled conditions. With controlled labeling and lighting/positioning variability, this tests basic detector capability to learn from synthetic data alone. The results in table 3 show YOLOv5 strongly outperforms Faster R-CNN when trained solely on synthetic images, achieving higher precision, recall and faster inference speed. YOLOv5's superior accuracy on pristine synthetic data suggests its one-stage architecture can more easily exploit the regularities of computer-generated images. It also trains 3x faster by sharing weights across the full image.

Table 3. Performance of object detectors trained with synthetic data.

Metric	YOLOv5	Faster R-CNN
$AP_{0.50:0.95}$	82,3 %	70,7 %
AP_{50}	97,8 %	93,5 %
Recall	96,3 %	85,6 %
FPS	49,75	25,67
Training time	03:08:15	09:00:30

4.2. Trained with complete dataset

Configuration 7 provides insight into model performance with the full diverse dataset, combining real and synthetic images for comprehensive variability. The results in Table 4 reveal YOLOv5 again outperforms Faster R-CNN when trained on the complete dataset, maintaining advantages in precision, recall, speed and training time. However, metrics for both models decrease considerably from the synthetic-only configuration 1. This demonstrates the challenge of bridging the synthetic-to-real gap, even with a blend of data. The magnitude of YOLOv5's precision and recall lead grows compared to earlier configurations. This implies YOLOv5 is better able to leverage diverse real and synthetic data for improved robustness. Faster R-CNN's greater reliance on "perfect" proposals likely

hinders adapting to real image nuances. Nonetheless, both models achieve viable detection capabilities on this task by complementing synthetic images with real-world samples.

Table 4. Performance of object detectors trained with complete dataset.

Metric	YOLOv5	Faster R-CNN
AP _{0.50:0.95}	81,6 %	66,8 %
AP ₅₀	97,2 %	91,7 %
Recall	95,5 %	82,6 %
FPS	58,8	22,66
Training time	04:04:28	11:46:59

4.3. Trained with classification dataset

Configuration 4 provides an isolated test of classification proficiency using cropped screw images, removing the localization component. This evaluates each model's screw type categorization capabilities in ideal, unobstructed settings. The results in Table 5 reveal nearly identical recall between YOLOv5 and Faster R-CNN, indicating comparable classification completeness when screw heads are centered and clearly visible. However, Faster R-CNN achieves slightly higher precision, suggesting its second stage helps refine classifications from the region proposal stage. Nonetheless, YOLOv5 retains considerable advantages in inference speed and training time due to its one-stage architecture. These computational efficiencies persist even without localization requirements.

Table 5. Performance of object detectors by classification.

Metric	YOLOv5	Faster R-CNN
AP _{0.50:0.95}	99,5 %	100 %
AP ₅₀	99,5 %	100 %
Recall	100 %	100 %
FPS	130,85	9,99
Training time	02:49:34	08:56:01

4.4. Trained with varying resolutions

Configurations 2, 3 and 7 provide insight into how image resolution impacts detection performance for YOLOv5 and Faster R-CNN. Lower resolution imagery contains less visual information but allows for faster processing. Higher resolutions enable capturing finer details at the cost of speed. As expected, inference times for both models improve significantly as resolution drops from 960 to 640 pixels, with 3-4x speedup. However, higher resolution delivers noticeably improved precision and recall for both detectors, despite slower speeds. Precision and recall are presented over dataset resolution in figure 3. This indicates the extra visual information facilitates detecting smaller, partially obstructed or ambiguous screws.

The training times ratio remained as in subsections 4.1 to 4.3 where YOLOv5 delivers approximately one third of the training

time from Faster R-CNN. In addition, both detectors show a similar dependence on resolution values.



Fig. 3. Performance of object detectors by dataset's resolution

4.5. Trained with varying dataset sizes

Configurations 5, 6, and 7 analyze detection performance as a function of training dataset size. More training data typically improves model generalization, but requires greater training time. Inference speed remains nearly constant regardless of training set size for both models. This indicates the trained model complexity does not substantially change with more data. However, as expected, larger training sets produce higher precision and recall - especially notable for rare classes. Faster R-CNN achieves slightly higher recall than YOLOv5 on the smaller data regimes, potentially benefiting from its region proposal localization. But with sufficient data, YOLOv5 matches or exceeds Faster R-CNN across metrics. This suggests YOLOv5 scales better with data volume. The similar precision growth versus dataset size implies both models can effectively leverage additional training data. But YOLOv5's faster training time indicates it assimilates new data more efficiently. Precision and recall are presented over dataset size in figure 4.



Fig. 4. Performance of object detectors by dataset's size

4.6. Composite interpretation

The composite results across diverse data configurations provide valuable insights into the comparative capabilities of YOLOv5 and Faster R-CNN for real-world screw detection. YOLOv5 consistently demonstrated superior precision, recall, speed and training efficiency - especially evident with blended synthetic and real data. This indicates YOLOv5's one-stage unified architecture is particularly well-suited for learning from synthesized data and generalizing to real-world images. Faster R-CNN achieved mildly better recall with limited data, potentially benefiting from its region proposal localization. However, YOLOv5 matched or exceeded Faster R-CNN's performance given sufficient training data. For cropped screw classification, both models performed similarly, with a slight precision advantage for two-stage Faster R-CNN. Overall, YOLOv5 proved the superior choice for real-world screw detection across most practical contexts and data conditions. Its efficiency with diverse data makes it an ideal detector for applications like waste electrical and electronic equipment recycling.

5. Conclusion and Future Work

This comparative study provides insights into one-stage vs. two-stage detectors for real-world screw detection applications. Across diverse configurations, YOLOv5 consistently surpasses Faster R-CNN in precision, recall, speed and training efficiency. YOLOv5's unified architecture appears better suited for learning from simulated and real image blends. Its single-shot predictions allow efficiently processing high resolution data. Faster R-CNN has minor advantages in limited data regimes, likely from its region proposal localization.

The next step of this project would be to implement a YOLO-based screw head detector to localize and classify screws on WEEE. Based on this detection an unscrewing system could be developed.

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